

Weekly Outline

- Readings refer to chapters in the [course textbook \(COD\)](https://canvas.wisc.edu/courses/485453/pages/required-course-textbook_) [_](https://canvas.wisc.edu/courses/485453/pages/required-course-textbook_) https://canvas.wisc.edu/courses/485453/pages/required-course-textbook_, unless otherwise specified.
- The full playlist for the online lectures can be found [here](https://www.youtube.com/playlist?list=PLYPFPSvBJgaeS8NVVMlmt_w24vUaWGI2C). [↗](https://www.youtube.com/playlist?list=PLYPFPSvBJgaeS8NVVMlmt_w24vUaWGI2C) [_](https://www.youtube.com/playlist?list=PLYPFPSvBJgaeS8NVVMlmt_w24vUaWGI2C) https://www.youtube.com/playlist?list=PLYPFPSvBJgaeS8NVVMlmt_w24vUaWGI2C Feel free to watch at faster playback speeds. The online lectures have been prepared by [Prof. Joshua San Miguel](https://directory.engr.wisc.edu/ece/Faculty/San-miguel_Joshua/). [_](https://directory.engr.wisc.edu/ece/Faculty/San-miguel_Joshua/) https://directory.engr.wisc.edu/ece/Faculty/San-miguel_Joshua/

You must read the assigned reading and watch the lectures before the Monday session.

Mon: M, Tue: T, Wed: W, Thu: R, Fri: F, Sat: Sa, Sun: Su

Week	Topics	Readings	Online Lectures
1	Introduction to Comp Arch	1.1-1.5	Lecture 1 ↗ _ https://www.youtube.com/watch?v=0o9gbdNjQyY
2	ISA Basics, ISA Advanced	2.1-2.4, 2.6-2.7, 2.5, 2.8-2.10	Lecture 2 ↗ _ https://youtu.be/_XWENd5si=VMZV3agSA12IPeU4 Lecture 3 ↗ _ https://youtu.be/_NCcEa6Isi=TV2shCA8jNW57V37

3	Performance Metrics, Arithmetic Logic Units	1.6-1.9, 3.1-3.2, B.5-B.6	Lecture 4  (https://youtu.be/Eftfzplido:si=tRa0cixl81Sos3w8)  (https://www.youtube.com/v=Bp5HLgYzxJo) Lecture 5  (https://www.youtube.com/v=Bp5HLgYzxJo)
4	Single-Cycle Control Path	4.1-4.4	Lecture 6  (https://www.youtube.com/v=cxJpfILlI4g) Lecture 7  (https://www.youtube.com/v=2BdW-tOIBoM)
5	Pipelining, Data Hazards	4.6-4.8	Lecture 8  (https://www.youtube.com/v=2Qj4PF6xUFU) Lecture 9  (https://www.youtube.com/v=vySekpWWRoE)

6	Control Hazards and Exceptions	4.9-4.10	Lecture 10  (https://www.youtube.com/watch?v=fopyBizSz2E)
7			

8	Cache Concepts, Cache Performance	5.1-5.4, 5.8	Lecture 11  (https://www.youtube.com/watch?v=zA2naghT0IQ) Lecture 12  (https://www.youtube.com/watch?v=bzBCHtVneAU)
9	Virtual Memory, Memory, ECC and I/O	5.6-5.7, B.9, 5.2, 5.11  (https://www.elsevier.com/_data/assets/pdf_file/0007/1092445/CH05A_Secured.pdf) , SECDED (http://jsm.ece.wisc.edu/ece552/secded.pdf)	Lecture 13  (https://www.youtube.com/watch?v=XjBkgV7C7PE) Lecture 14  (https://www.youtube.com/watch?v=JxooySjoh5c)
10	Superscalar and Out-of-Order, MIPS R10K	4.11, MIPSR10K (http://jsm.ece.wisc.edu/ece552/mipsr10000.pdf)	Lecture 15  (https://www.youtube.com/watch?v=Yt-ts9gWgUo) Lecture 16  (https://www.youtube.com/watch?v=x9so8ZIOYA4)

11	❄️❄️❄️🌸 Spring Break 🌸❄️❄️❄️		
12	Parallel Processors, Shared Memory and Coherence	6.1-6.2, 6.4-6.5 5.9-5.10	Lecture 17 📄 v=VnpVoZy5ual) Lecture 18 📄 v=IHp5euguz4k)
13	Consistency and SIMD	6.3, 6.6	Lecture 19 📄 v=PihWN1vrqdM)
14	Integer Multiplication/Division	3.3-3.5	Lecture 20 📄 v=

	Floating Point		v=_yIiVnpsGA8) Lecture 21  https://www.youtube.com/watch?v=sZLqA6x6_ws
15			
16			

